

Tushar Jayanti

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SUMMARY

Senior Backend Engineer with 7+ years building high-availability distributed systems and production SaaS platforms. Experienced in designing secure, multi-tenant backend services, high-performance APIs, and LLM-powered agentic systems. Currently building production agentic AI applications using **FastAPI, Python, and Claude APIs**. Strong focus on scalable architecture, observability, and shipping reliable systems in fast-paced environments.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C#, JavaScript

Backend: FastAPI, Flask, Spring Boot, gRPC, REST, GraphQL, Kafka, Microservices, Event-Driven Architecture

Databases: PostgreSQL, MSSQL, MongoDB, Redis, Elasticsearch, DynamoDB

Cloud/DevOps: AWS, Docker, Kubernetes, Terraform, GitLab CI, Jenkins, Datadog

Data & Orchestration: Apache Airflow, Batch Processing, Real-time Event Systems

AI/LLM: LLM API integration (Anthropic Claude), agentic tool orchestration, pgvector, hallucination-detection eval loops, prompt engineering, Claude Code, GitHub Copilot

EXPERIENCE

DISCO

Senior Software Engineer

Nov 2023 – Sept 2025

Platform / Identity Systems

- Led migration of mission-critical identity platform from .NET + RavenDB to **Kotlin + Spring Boot + PostgreSQL**, improving scalability and long-term maintainability.
- Built and owned **multi-tenant, permissions-aware identity and authorization services** with audit logging, supporting enterprise-grade access control across thousands of customers.
- Designed and owned high-performance backend APIs and service-to-service communication using **Python (FastAPI/Flask) and gRPC with Protocol Buffers**.
- Migrated Python authorization service from REST to **gRPC with Protocol Buffers**, enabling strongly typed contracts and high-performance inter-service communication.
- Reduced **p99 latency from 4.2s to 1s (75%+ improvement)**, improving throughput and efficiency under production load.
- Built custom **LINQ-style C# → PostgreSQL query framework**, improving data access safety, consistency, and developer productivity.
- Improved **CI/CD speed by 67%** and reduced Docker image sizes by migrating **Python services from pipenv to uv**.
- Provisioned and managed infrastructure using **Terraform + AWS**, improving deployment consistency and system reliability.
- Mentored engineers and drove architectural best practices across platform services.

PurpleToko

Founding Engineer

Oct 2022 – Oct 2023

Distributed E-commerce Platform

- Built and iterated backend APIs and systems on AWS for search, order processing, and payments in a fast-paced startup environment.
- Implemented **Elasticsearch-based search** and **SQS-driven asynchronous pipelines** for resilient distributed processing.
- Designed stateless services enabling horizontal scaling and fault isolation in a 0→1 growth environment.
- Translated ambiguous product requirements into scalable backend systems and production-ready services.

Transcend Street Solutions

Software Engineer

Sept 2020 – Sept 2022

Financial Systems

- Engineered and launched the **Reserve Release** feature end-to-end, preventing incorrect financial orders across **10k+ daily transactions**.
- Optimized SQL workloads, reducing latency **8x (6s to 700ms)** through indexing and query optimization.
- Built and orchestrated batch data pipelines using **Apache Airflow**, enabling reliable and observable workflow execution.
- Migrated monolith modules to **Spring Boot microservices** and built REST APIs for high-throughput workflows.

Baanyan Software Services (USAA)

Full Stack Java Developer

Aug 2018 – Jul 2020

Financial Systems

- Developed Spring Boot services integrated with **Kafka-based event-driven architecture** for high-throughput financial processing.

- Created multiple plugins using Python to extend functionality for third-party tools.
- Built monitoring dashboards using **Prometheus and Grafana**, improving observability and system reliability.

PROJECTS

vox-agent

github.com/tusharjayanti/vox-agent

- Built a production-grade **agentic customer support AI** using **FastAPI, asyncpg, and pgvector**, with an inline **hallucination-detection eval loop** using Claude Sonnet as generator and Claude Haiku as LLM judge.
- Designed end-to-end backend architecture: async API layer, vector similarity retrieval, and structured LLM orchestration with tool use.

calculator-agent

github.com/tusharjayanti/calculator-agent

- Implemented an **agentic tool-use pipeline** using the Anthropic API, demonstrating structured multi-step reasoning and tool orchestration patterns.

Intelligent Chatbot

github.com/tusharjayanti/TensorflowChatbot

- Built a **seq2seq chatbot** using **Python and TensorFlow**, trained on a movie dialogue dataset to generate contextually relevant responses.

Optical Character Recognition

github.com/tusharjayanti/OpticalCharacterRecognition

- Developed an **OCR system** using **Linear and Logistic Regression in Python and TensorFlow** to improve accuracy on small fonts and noisy image data for CAPTCHA recognition.

EDUCATION

Master of Science in Computer Science — New Jersey Institute of Technology

Bachelor of Technology in Electrical Engineering — Guru Gobind Singh Indraprastha University